

A close-up photograph of a yellow pencil with a sharpened lead tip, resting on a stack of white papers. The papers have faint blue horizontal lines. The word 'Telnet' is printed in a dark blue, serif font, centered on one of the pages. The background is softly blurred, showing more of the paper stack and a hint of a blue object at the top.

Telnet

Remote Logging

- ✿ It would be impossible to write a specific client/server program for each demand.
- ✿ The better solution is a general-purpose client/server program that lets a user access any application program on a remote computer.
- ✿ TELNET is a general-purpose client/server application program.

Telnet

- ✿ Timesharing – UNIX

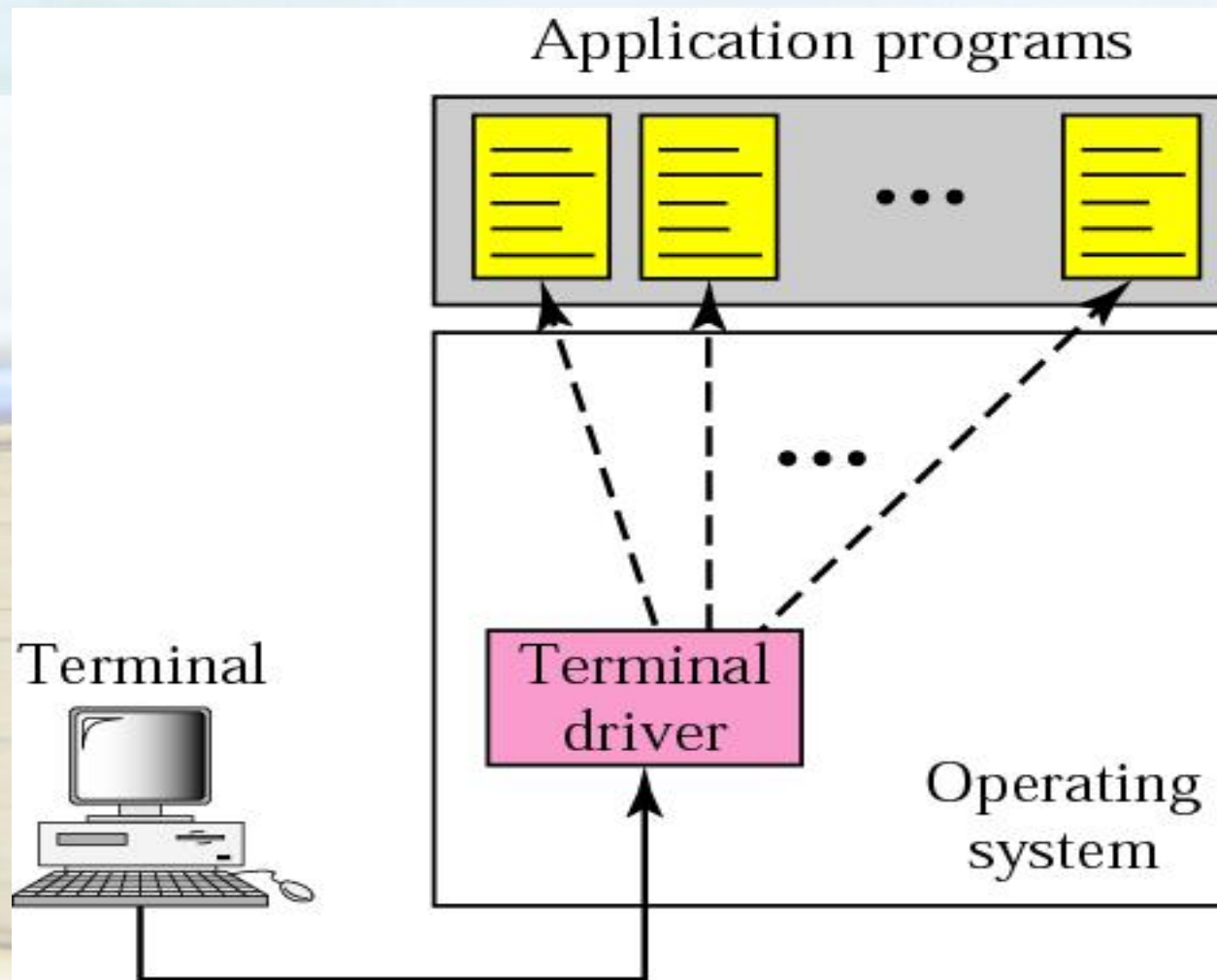
- ✿ Logging

 - ✿ Local Log –in

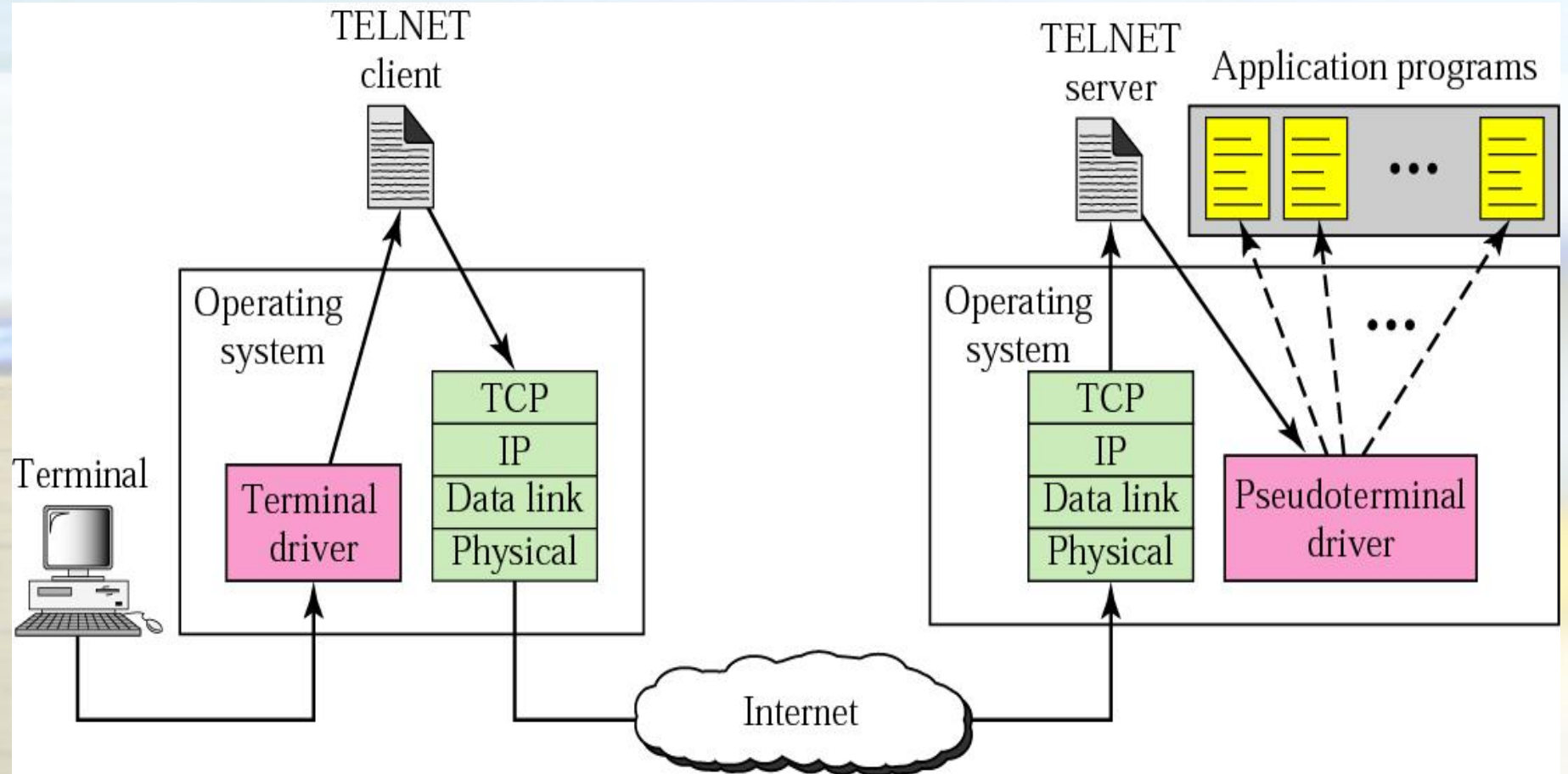
 - ✿ Remote log –in

 - ✿ Needs a conversion from local character set to universal character set (NVT – Network Virtual Terminal)

Local Log –in



Remote log-in



Network Virtual Terminal

- ✿ End of file

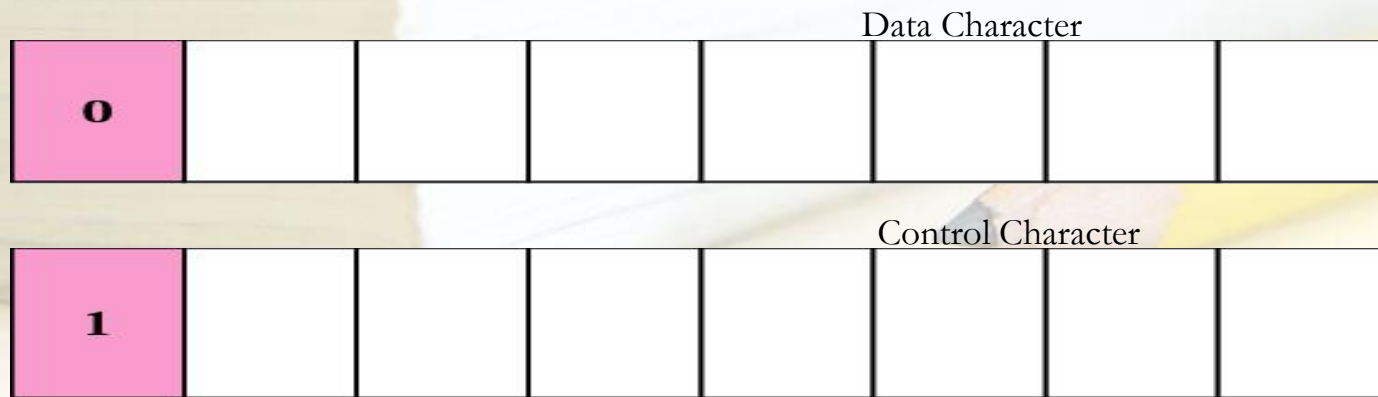
- ✿ DOS – Ctrl+z

- ✿ UNIX - Ctrl+d

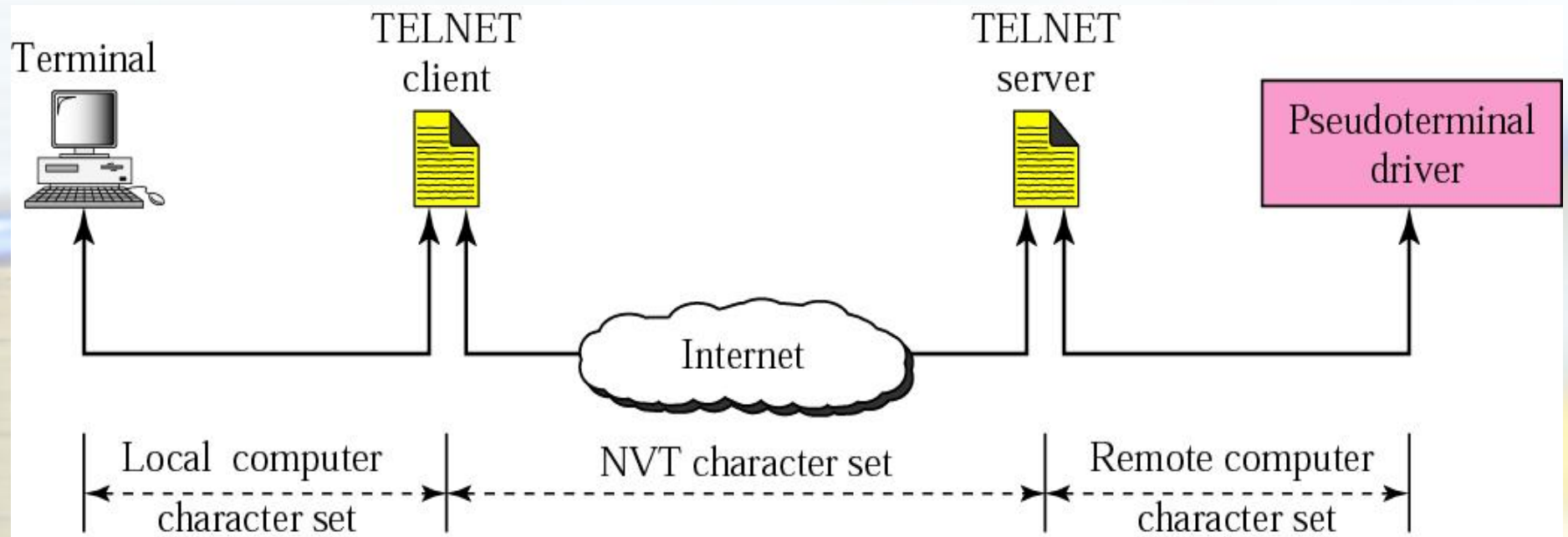
- ✿ NVT Character Set

- ✿ 8 bit byte

- ✿ NVT uses two sets of characters, one for data and one for control



Concept of NVT



Some NVT control characters

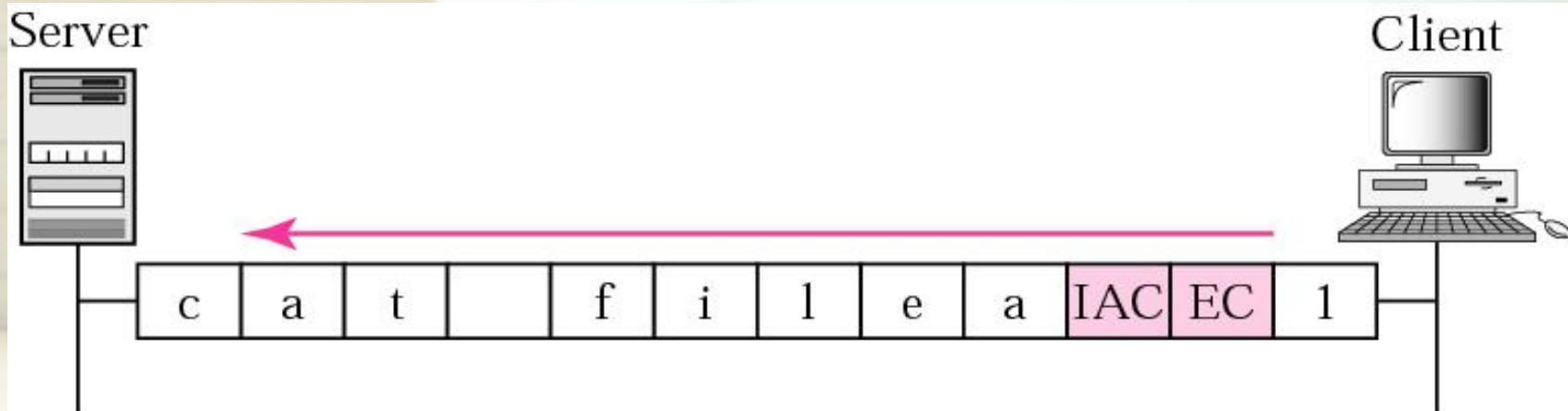
<i>Character</i>	<i>Decimal</i>	<i>Binary</i>	<i>Meaning</i>
EOF	236	11101100	End of file
EOR	239	11101111	End of record
SE	240	11110000	Suboption end
NOP	241	11110001	No operation
DM	242	11110010	Data mark
BRK	243	11110011	Break
IP	244	11110100	Interrupt process
AO	245	11110101	Abort output
AYT	246	11110110	Are you there?
EC	247	11110111	Erase character
EL	248	11111000	Erase line
GA	249	11111001	Go ahead
SB	250	11111010	Suboption begin
WILL	251	11111011	Agreement to enable option
WONT	252	11111100	Refusal to enable option
DO	253	11111101	Approval to option request
DONT	254	11111110	Denial of option request
IAC	255	11111111	Interpret (the next character) as control

Embedding

- ✿ The same connection is used by TELNET for sending both data and control characters.
- ✿ TELNET accomplishes this by embedding the control characters in the data stream.

✿ *cat file1*

✿ *cat filea<backspace>1*



Options

- ✿ TELNET lets the client and server negotiate options before or during the use of the service.
- ✿ Options are extra features available to a user with a more sophisticated terminal.

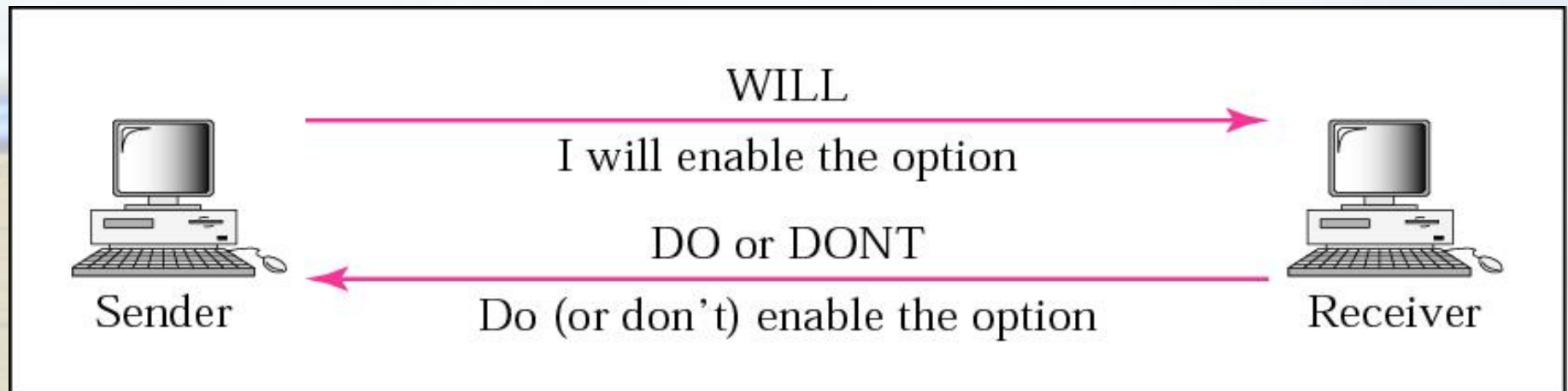
<i>Code</i>	<i>Option</i>	<i>Meaning</i>
0	Binary	Interpret as 8-bit binary transmission
1	Echo	Echo the data received on one side to the other
3	Suppress go ahead	Suppress go-ahead signals after data
5	Status	Request the status of TELNET
6	Timing mark	Define the timing marks
24	Terminal type	Set the terminal type
32	Terminal speed	Set the terminal speed
34	Line mode	Change to line mode

Option Negotiation

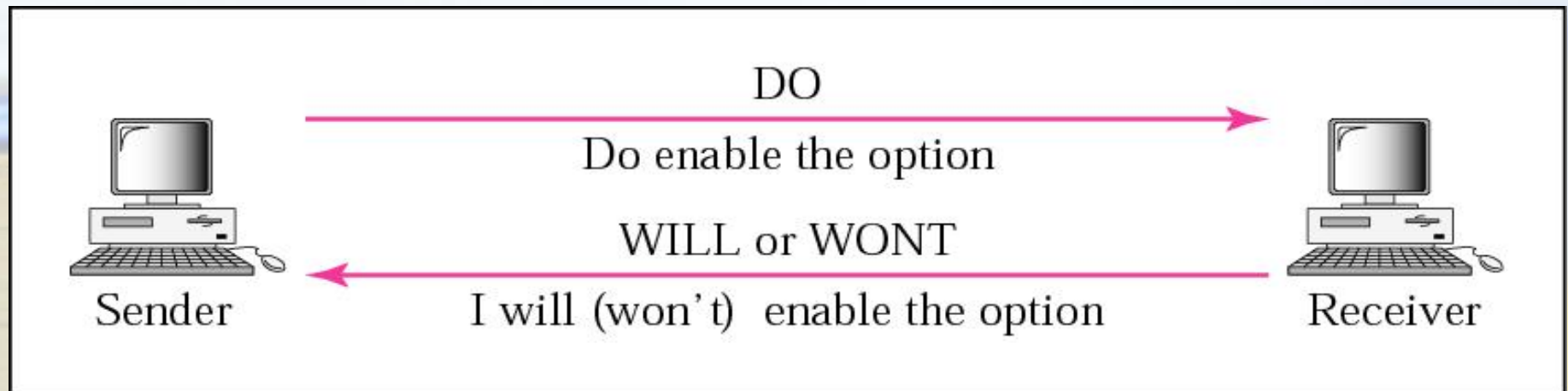
- ✿ To use any of the options first requires option negotiation between the client and the server.
- ✿ Four control characters are used for this purpose.

<i>Character</i>	<i>Decimal</i>	<i>Binary</i>	<i>Meaning</i>
WILL	251	11111011	1. Offering to enable 2. Accepting a request to enable
WONT	252	11111100	1. Rejecting a request to enable 2. Offering to disable 3. Accepting a request to disable
DO	253	11111101	1. Approving an offer to enable 2. Requesting to enable
DONT	254	11111110	1. Disapproving an offer to enable 2. Approving an offer to disable 3. Requesting to disable

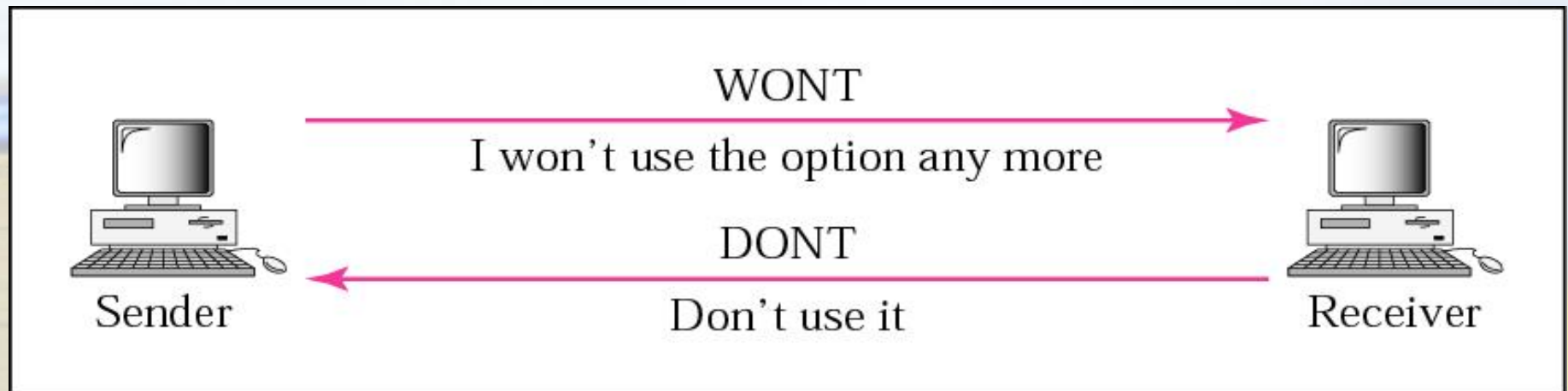
Offer to enable an option



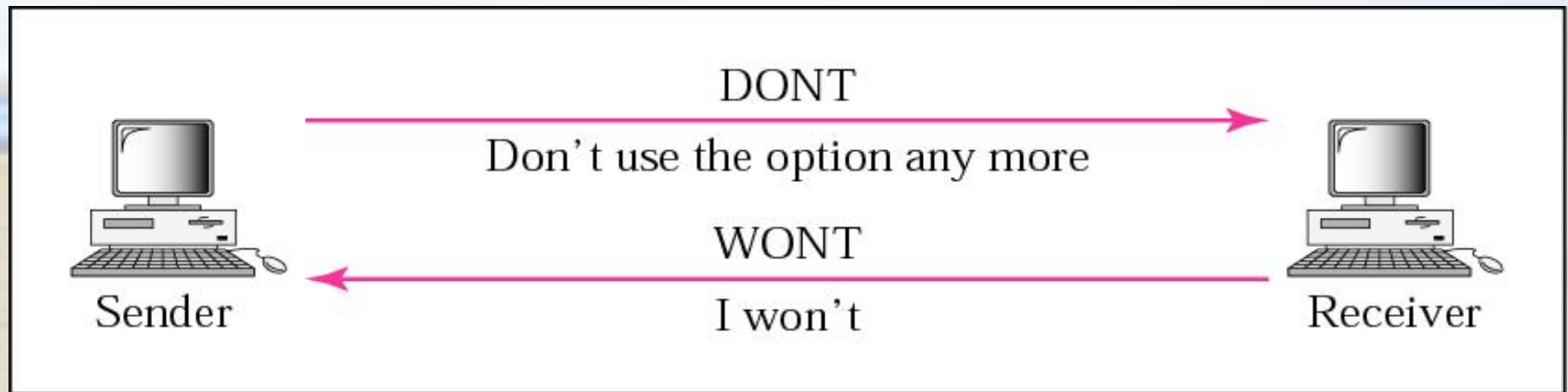
Request to enable an option



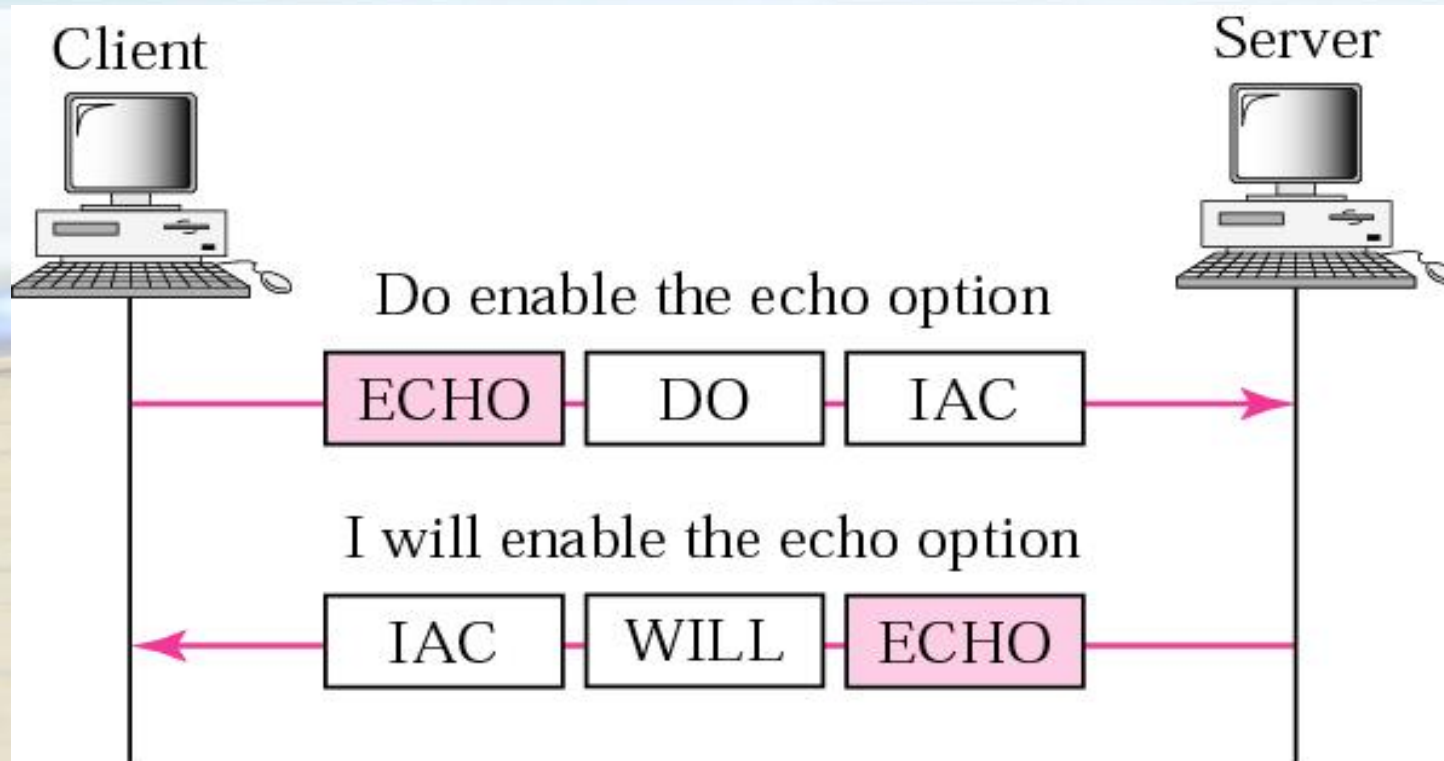
Offer to disable an option



Request to disable an option



Echo option

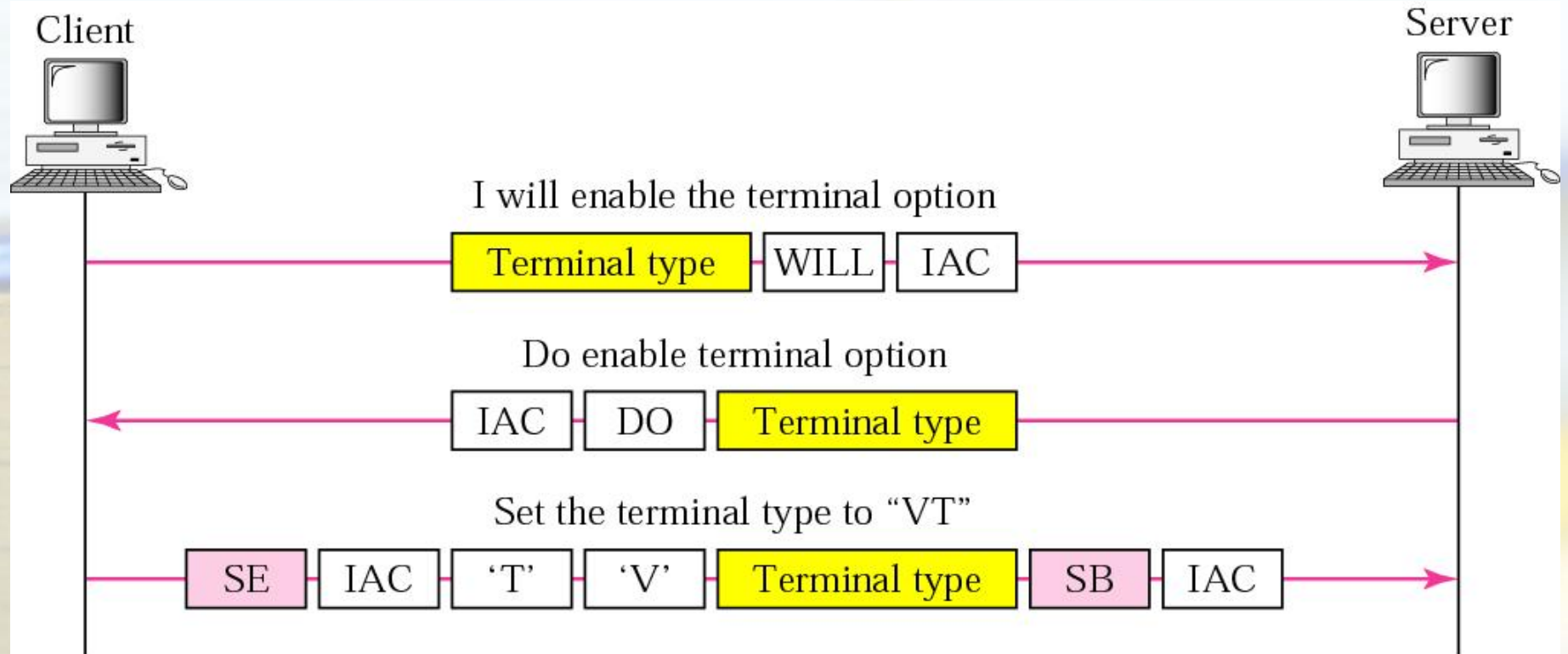


Suboption Negotiation

- ✿ Some options require additional information.
- ✿ The client and server agree through a process called suboption negotiation.

<i>Character</i>	<i>Decimal</i>	<i>Binary</i>	<i>Meaning</i>
SE	240	11110000	Suboption end
SB	250	11111010	Suboption begin

Example of suboption negotiation



Mode of Operation

✿ Default mode

✿ Character Mode

✿ Line Mode